Portable Spec Kit

AI Agentic Specification-Driven Development

A single portable file that defines your complete development methodology — from specifications to deployment. Drop it into any project and the AI agent follows your engineering standards, manages tasks, writes tests, and maintains context across sessions.

SPECS → PLANNING → TASKS → BUILD + TEST → TRACK

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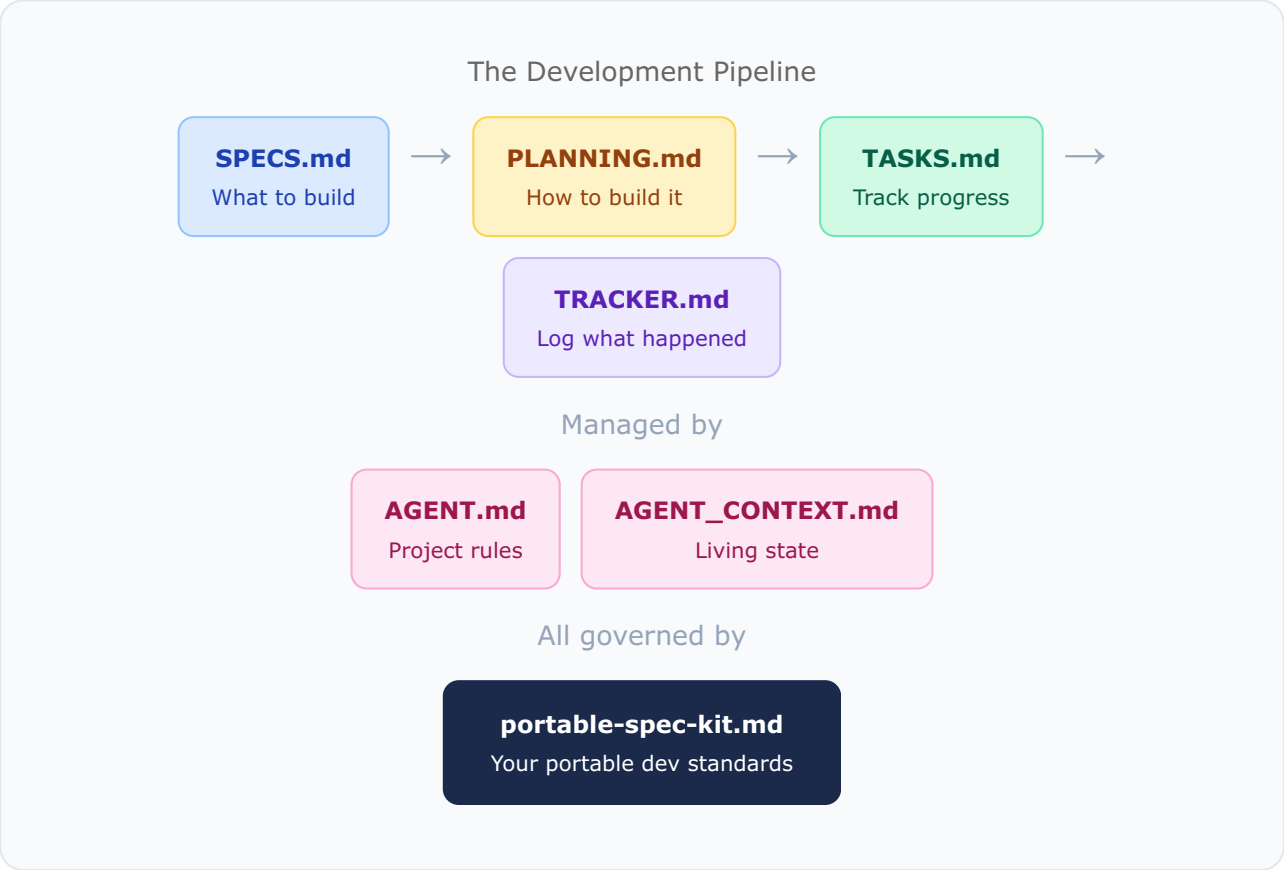
The Framework

The **Spec-Driven Development Framework** is a methodology where every project follows a structured pipeline from requirements to deployment, managed by an AI agent that enforces your personal engineering standards.



8 Core Strengths

Strength	What It Means
🪶 **Lightweight**	Single markdown file. Zero dependencies. Zero install.
📦 **Portable**	One file → any repo. Works instantly. Symlinks for all agents.
🤖 **Agent-Agnostic**	Claude, Copilot, Cursor, Windsurf, Cline — one source, all sync.
🏗️ **Self-Scaffolding**	Auto-creates 6 agent files, README, workspace context. Project structure in seconds.
📋 **Spec-Driven**	SPECS → PLAN → TASKS → TRACK. Complete dev pipeline. Retroactive spec-filling.

🧠 **Context-Persistent**	AGENT_CONTEXT.md carries state across sessions. Pick up after weeks. Never lose context.
🔄 **Non-Blocking**	Code first, specs later. Agent tracks silently. Never blocks the developer.
✅ **Self-Validating**	Agent tests everything. Fixes before presenting. 98%+ coverage target.

What the Framework File Contains

The `portable-spec-kit.md` file is the single source of truth for your development methodology. It contains:

Section	What It Governs
Git & GitHub Rules	When to commit, push, ask before critical ops
Security Rules	Secret handling, .env practices, code security
Versioning	v0.1 → v1.0 numbering, changelog standards
Task Tracking	Add tasks FIRST, then work. Module-based organization.
Testing Standards	Coverage targets, edge case checklist, mock rules, self-validation
Code Quality	Review checklist, naming conventions, deployment checklist
Error Handling	Structured errors, logging, error boundaries, user-friendly messages
Branch & PR Workflow	Feature branches, PR format, squash merge, clean history
Dependencies	Bundle size checks, lock files, audit, avoid unnecessary deps
Project Templates	6 agent files + README auto-created with standard structure
Auto-Scan	Detects existing projects, creates missing files, restructures to standard
Agent Behavior	Guide don't enforce, silent tracking, retroactive spec-filling

Multi-Agent Support

One source file — `portable-spec-kit.md` — works with every AI coding agent via symlinks:

Agent	File Created	How
Claude Code	`CLAUDE.md`	Symlink → portable-spec-kit.md
GitHub Copilot	`.github/copilot-instructions.md`	Symlink → portable-spec-kit.md
Cursor	`.cursorrules`	Symlink → portable-spec-kit.md
Windsurf	`.windsurfrules`	Symlink → portable-spec-kit.md
Cline	`.clinerules`	Symlink → portable-spec-kit.md

macOS/Linux: Uses symlinks — edit `portable-spec-kit.md`, all agents read the update instantly.

Windows: Uses copies — re-run install command after editing, or let the AI agent keep files in sync.

Complete Flow

1. **Install** — one command downloads `portable-spec-kit.md` + creates symlinks for all agents (Claude, Copilot, Cursor, Windsurf, Cline)

↓

2. **Open any AI agent — auto-scaffolds** — detects environment, creates WORKSPACE_CONTEXT.md, creates `agent/` with 6 files (AGENT, SPECS, PLANNING, TASKS, TRACKER, CONTEXT), project dirs (src/, tests/, docs/), README, .gitignore, .env.example

↓

3. **Work however you want** — "build me X" → agent adds to TASKS.md, builds, tests. "What should I do next?" → agent walks through spec-driven flow. Never blocks.

↓

4. **Session ends** — agent updates AGENT_CONTEXT.md with progress, decisions, what's next

↓

5. **Come back later** — days, weeks, months. Agent reads AGENT_CONTEXT.md: "Here's where we left off." Continues exactly where you stopped.

↓

6. **Edit the framework** — update portable-spec-kit.md, all 5 agent symlinks read the change instantly. Standards evolve with you.

Project Structure

```
your-project/
├── portable-spec-kit.md   ← Framework file (source)
├── CLAUDE.md              ← Symlink (Claude Code)
├── .cursorrules           ← Symlink (Cursor)
├── WORKSPACE_CONTEXT.md      ← Auto-created (workspace state)
├── README.md              ← Auto-created (standard structure)
│
├── agent/                 ← Auto-created (project management)
│   ├── AGENT.md           ← Project rules, stack, brand
│   ├── AGENT_CONTEXT.md   ← Living state (updated every session)
│   ├── SPECS.md           ← Requirements & features
│   ├── PLANNING.md        ← Architecture, methodology & research
```

```
│   ├── TASKS.md          ← Task tracking (module-based)
│   └── TRACKER.md        ← Version history
│
├── src/                  ← Your code
├── tests/                ← Your tests
├── ard/                  ← Architecture docs (HTML + PDF)
└── ...
```

The 6 Agent Files

File	Purpose	When
`AGENT.md`	Project-specific instructions — stack, rules, brand, AI config	Setup
`AGENT_CONTEXT.md`	Living state — done, next, decisions, blockers, file structure	Every session
`SPECS.md`	What to build — requirements, features, acceptance criteria, scope	Before dev
`PLANNING.md`	How to build — architecture, data model, phases, methodology, research, decision log	Before dev
`TASKS.md`	Module-based task tracking with progress summary table	During dev
`TRACKER.md`	Version changelog (categorized), test results, deployment log	End of version

How to Work in Spec-Driven Development

Phase 1: Specification

> **Goal:** Define WHAT you're building before touching code.

Start every project by writing `agent/SPECS.md`. This is a conversation between you and the AI:

You Say	AI Does
"I want to build a resume editor"	Writes requirements, features table, scope boundaries, acceptance criteria in SPECS.md

"Add AI-powered bullet rewriting"	Adds feature to specs, estimates priority, updates scope
"That's out of scope for v1"	Moves to "Out of scope (future)" section

Rule: No code is written until SPECS.md has requirements and acceptance criteria.

Phase 2: Planning

> **Goal:** Define HOW you're building it — architecture, data model, phases.

AI writes `agent/PLANNING.md` based on approved specs:

Section	What AI Fills In
Stack	Recommends tech (frontend, backend, DB, hosting) with trade-offs. You approve.
Architecture	System design, component relationships, data flow
Directory Structure	Exact file/folder layout for the chosen stack
Data Model	Tables, types, relationships
API Endpoints	Method, path, description for each route
Build Phases	Phased delivery plan — foundation first, features later
Security	Auth, CORS, input validation, secret handling

Rule: Deployment is deferred to release time. Don't discuss hosting during planning.

Phase 3: Execution

> **Goal:** Build, test, and track every task.

AI creates `agent/TASKS.md` with module-based breakdown:

```
## Module 1: Project Setup
| # | Task | Status |
|---|------|:------:|
| 1.1 | Initialize Next.js + Tailwind | [x] |
| 1.2 | Create component structure | [ ] |
```

```
## Module 2: Core Features
| # | Task | Status |
|---|------|:------:|
| 2.1 | Build editor canvas | [ ] |
```

Execution rules enforced by the framework:

Rule	What It Means
Tasks first	When you ask for something, AI adds it to TASKS.md BEFORE starting work
Self-validate	AI runs tests, fixes failures, only presents stable results
Test everything	Every feature gets tests — unit, integration, UI interaction
Edge cases	Empty inputs, boundaries, XSS, unicode, error paths — all tested
Coverage targets	Backend 98%+, frontend logic 98%+, UI components 85%+
No broken features	User should NEVER discover something doesn't work
Context updated	AGENT_CONTEXT.md updated at end of every session

Phase 4: Tracking

> **Goal:** Log what was delivered, tested, and deployed.

At end of each version, AI updates `agent/TRACKER.md` :

```
## v0.2 — AI Features (March 30, 2026)

### Summary
Added JD-tailored resume, PDF generation, 604 tests.

### Changes
- **Frontend:** JD Tailor modal, version manager, skills tags
- **Backend:** 3 PDF engines, expand API, fetch-url API
- **AI:** Bullet format detection, deduplication, length matching

### Tests
- 604 tests passing, 95% coverage

### Deployment
```

```
- Deployed to: aqibmumtaz.github.io
- Date: 2026-03-30
```

Daily Workflow Summary

1. **Start session** — AI reads agent/AGENT.md + AGENT_CONTEXT.md. Knows exactly where you left off.

↓

2. **You give instructions** — "add dark mode", "fix scoring bug", "build PDF export"

↓

3. **AI adds to TASKS.md** first, then builds, tests, and shows results with coverage

↓

4. **You review** — approve, adjust, or ask for changes. AI fixes and re-tests.

↓

5. **You say "commit"** — AI commits with descriptive message. "push" sends to remote.

↓

6. **Session ends** — AI updates AGENT_CONTEXT.md. Next session picks up seamlessly.

> **The result:** Every project follows the same spec → plan → build → test → track pipeline. Nothing is ad-hoc. Nothing falls through cracks. Context never lost.

Agent Behavior — Guide, Don't Enforce

The framework is **not rigid**. The AI agent adapts to how you want to work:

> **Core principle:** The user can start anywhere — jump into coding, give direct tasks, ask questions, or follow the full spec-driven flow. All valid. The agent adapts to the user, not the other way around.

You Do	Agent Does
"Build me a login page"	Adds to TASKS.md → builds it → tests it → shows results
"Fix this bug"	Adds to TASKS.md → fixes → marks done
"What should I do next?"	Walks through spec-driven process: define specs → plan → tasks
"What's the status?"	Shows from TASKS.md + AGENT_CONTEXT.md
Come back after weeks	Reads AGENT_CONTEXT.md → summarizes where you left off
Never wrote specs	Retroactively fills SPECS.md from what's built — no complaints

Silent Tracking

Even if you never mention the process, the agent:

- Adds every task to TASKS.md before starting
- Marks tasks done when complete
- Updates AGENT_CONTEXT.md at session end
- Fills gaps in SPECS.md and PLANNING.md retroactively
- Surfaces process naturally: "I've added this to TASKS.md", "Based on SPECS.md, we still have..."

> **User's time is sacred.** Agent does 90% of the work — writes specs, plans, tasks, tests, docs. User reviews 10%. Never ask the user to write specs or plans — the agent drafts everything, user approves or adjusts.

File Creation/Update Rule

This single rule governs how the agent handles ALL managed files (`WORKSPACE_CONTEXT.md`, `README.md`, and all `agent/` files):

Scenario	Action
File does not exist	Create from standard template, fill in known details
File exists but wrong structure	Restructure to match template — **retain all existing content**, never lose data
File already matches template	Leave as-is

> **Never lose content.** When restructuring an existing file, every detail, decision, and note is preserved — just reorganized into the standard sections.

Customization

About the User — Why It Matters

The "About the User" section tells the AI agent **who it's working with**. This directly affects how the agent behaves:

What You Set	How the Agent Adapts
Your expertise (e.g., "PhD in AI" vs "first-time coder")	Adjusts technical depth, skips or explains concepts
Communication style (e.g., "direct, data-driven")	Uses tables, evidence, concise language vs. conversational
Working pattern (e.g., "iterative, starts brief")	Adapts planning granularity and suggestion style
AI delegation (e.g., "AI does 90%")	Controls how autonomous the agent is vs. asking for approval

Auto-populated on install from your GitHub profile (name, bio) via `setup.sh`. Customize further:

```
## About the User
- **Jane Smith** — Senior backend engineer, 8 years Python/Go
- Communication style: direct, prefers code examples over explanations
- Working pattern: TDD, small PRs, iterative
- AI delegation: AI does 80%, I review everything before commit
```

> **Fork-friendly + auto-populated.** On install, `setup.sh` pulls your name and bio from GitHub. Customize the rest manually. The public repo ships with a generic placeholder — no merge conflicts for forkers.

Your Projects

Project-specific details go in `agent/AGENT.md` (auto-created per project). This keeps the framework file portable while each project has its own stack, brand, and rules:

```
## Stack
| Layer | Technology |
|-------|-----------|
| Frontend | Next.js + TypeScript |
| Backend  | FastAPI |

## Brand
- Primary: #1B2A4A
- Accent: #4a6fa5
```

Dev Stack Adaptation

The framework uses stack-agnostic language for testing, building, and linting. Each project's `agent/AGENT.md` defines the actual commands:

Framework Says	Project Defines
"All tests passing"	`npm test` / `pytest` / `go test`
"Type checking: zero errors"	`npx tsc --noEmit` / `mypy`
"Linting: zero errors"	`eslint` / `ruff` / `golint`
"Build succeeds"	`npm run build` / `docker build`
"PDF generation tool"	WeasyPrint / Puppeteer / wkhtmltopdf

> **Stack-agnostic by design.** The framework never assumes a specific language, build tool, or test runner. It defines *what* to check — each project defines *how*.

Sharing & Syncing the Framework

Recommended: Dedicated GitHub Repo

📦 Single-Source-of-Truth Repo

One repo holds portable-spec-kit.md. All projects pull from it. Updates pushed back from any project.

```
# Setup (once)
mkdir portable-spec-kit && cd portable-spec-kit
git init && cp /path/to/portable-spec-kit.md .
git add . && git commit -m "Framework v1.0"
gh repo create portable-spec-kit --private --push
```

```
# Pull into any project
curl -sO https://raw.githubusercontent.com/you/portable-spec-kit/main/portable-spec-kit.md

# Push updates back from any project
cp portable-spec-kit.md ~/portable-spec-kit/
cd ~/portable-spec-kit && git add . && git commit -m "Updated from ProjectX" && git push
```

✓ Version controlled ✓ Pull anywhere ✓ Change history ✓ Cross-machine sync

Other Options

Method	How	Best For
Git Submodule	`git submodule add repo .framework`	Auto-sync across repos
GitHub Gist	`gh gist create portable-spec-kit.md`	Quick sharing, browser editable
Shell Alias	`alias init-dev='curl -sO URL/portable-spec-kit.md'`	One-command setup

Update Workflow Across Projects

1. **Discover new practice** while working on Project A (e.g., new testing rule)

↓

2. **AI updates portable-spec-kit.md** in Project A with the new rule

↓

3. **Push to shared repo** — single source of truth updated

↓

4. **Pull in Project B** — instantly has the new rule. AI follows it immediately.

Portable vs Project-Specific

In portable-spec-kit.md (Portable Framework)	In agent/AGENT.md (Per Project)
Git/push/commit rules	Tech stack choices
Security practices	Brand colors & fonts
Testing methodology	AI model/provider config
Naming conventions	Dev server port
Project templates & auto-scan	Deployment specifics
Code review standards	Project-specific rules
Spec-driven development flow	API endpoints & data model
Version numbering scheme	Database schema

Examples Included

The repo includes two example projects showing the framework in action:

Example	What It Shows	Best For
`examples/starter/`	Fresh project right after setup. Self-documenting README explains every file and directory.	Understanding the framework — **start here**
`examples/my-app/`	Mid-development Next.js + Supabase project. 11/16 tasks done, 24 tests at 92% coverage.	Seeing a realistic project in progress

Starter Project Structure

```
examples/starter/
├── portable-spec-kit.md   ← Framework file (source)
├── CLAUDE.md              ← Symlink (Claude Code)
├── .cursorrules           ← Symlink (Cursor)
├── WORKSPACE_CONTEXT.md   ← Auto-created workspace state
├── README.md              ← Self-documenting (explains everything)
├── agent/
│   ├── AGENT.md           ← Stack: TBD (waiting for your specs)
```

```
│   ├── AGENT_CONTEXT.md   ← Status: "Setup — waiting for specs"
│   ├── SPECS.md           ← Empty — ready for requirements
│   ├── PLANNING.md        ← Empty — ready for architecture & methodology
│   ├── TASKS.md           ← 1/5 tasks (project initialized)
│   └── TRACKER.md         ← v0.1 placeholder
```

Mid-Development Project

```
examples/my-app/
├── agent/
│   ├── AGENT.md           ← Next.js + Supabase configured
│   ├── AGENT_CONTEXT.md   ← v0.1: 11/16 tasks, 24 tests
│   ├── SPECS.md           ← 8 features, priorities, criteria
│   ├── PLANNING.md        ← Data model, APIs, 3 phases
│   ├── TASKS.md           ← 5 modules with progress
│   └── TRACKER.md         ← Categorized changelog
```

> **One file. Any project. Your standards.**
> The framework travels with you — consistent engineering practices across every project, every team, every AI session.

Portable Spec Kit

Thank You

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